
TMS-001100

VSEPR-SIM v5.0.0-beta.6 — Formation and Premacro Analysis Foundation

Release Report: Trajectory-Truth Analysis, Crystal Imperfections, Diffusion, and Macro Inference

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1 Abstract

VSEPR-SIM version 5 is the penultimate version iteration, finally touching on true macro-scale properties. Beta-6 fills in several mechanisms for lower-level verification before doing so — implications initially skipped for simplification in previous versions but impossible to ignore when moving onto very complex systems in coming developments (v5-beta.8).

Beta-6 is the **instrumentation release**. It does not claim to be macro. It claims to have built the instruments required before macro can be trusted.

The result is a working trajectory-to-inference pipeline that can measure structural change, surface interaction, diffusion behavior, and bead/powder packing from trajectory data — *without* storing derived material properties in simulation state. This distinction is the entire project philosophy in practice.

2 Version Identity

Version	Role	Description
v4.1.0	Stable foundation	Frozen known-good baseline
v5.0.0-beta.6	Instrumentation	Formation + premacro analysis foundations complete
v5.0.0-beta.7	Unification	Formation outputs feed analyzers and reports
v5.0.0-beta.8	Coupling	Analyzers feed macro-directed candidate workflows

Table 1: Version roadmap context.

Beta-6 was developed alongside v4.1.x but with entirely different goals. v4.1.x is a stability-preserving branch. v5-beta is an architecture-expanding branch. They do not conflict.

3 Core Design Rule — Preserved Across All New Systems

No inferred macro property is written back into State.

The system respects four clean layers:

Layer	Role
State	Position, velocity, identity, dynamic truth
xyzFull	Trajectory evidence
Analysis	Measures behavior from trajectory
Inference	Estimates macro behavior from measured behavior

Table 2: Strict four-layer architecture.

Quantities explicitly **not** stored in State include: diffusion coefficient, effective diffusivity by axis, transport class, porosity, bulk density, packing class, compressibility proxy, permeability proxy, sintering readiness, and formation energy proxy.

These are analysis-only outputs. There is no field in `atomistic::State` for any of them. That is not accidental. It is the architecture.

4 Measurement Backbone — Completed Components

Key validation result: Kabsch RMSD ignores rigid translation and rotation; it responds to true non-rigid deformation. Mean displacement captures raw motion. Structural residual

Component	Role
<code>vsepr::Vec3</code> authoritative state vector	Single canonical position/velocity type
Eigen bridge (isolated at analysis boundaries)	Matrix ops without contaminating core state
Kabsch / RMSD analysis	Rotation-invariant structural residual
Stationarity / settling gates	Convergence check before inference
Translation / rotation / deformation distinction	Separates motion types from structural change

Table 3: Beta-6 measurement infrastructure.

captures reference-site mismatch. This gives the project a defensible way to separate *moved*, *rotated*, *deformed*, *relaxed*, and *destabilized* — a distinction required before any macro inference can be credible.

5 Task 9 — Crystal Imperfection Emergence

PASS

Crystal imperfection detection validated across SC, FCC, and BCC lattices:

Variant	Detected behavior
ideal	Low RMSD, no occupancy mismatch, no excess atoms
interstitial	$N_{\text{excess}} = +1$
vacancy	$N_{\text{excess}} = -1$, occupancy mismatch
substitutional	<code>structure_class = ideal</code> , <code>identity_class = substitutional_site</code>
thermal_jitter	Displacement disorder, no false count defect
localized_impulse	Local relaxation / disruption
multi_site	Broad disorder / instability

Table 4: Crystal imperfection variants and detected behavior.

Primary achievement: Substitutional defects are separated into structural and identity channels. The analyzer correctly reports:

```
structure_class = ideal_crystal
identity_class  = substitutional_site
```

A substitution can be geometrically clean while identity-defective. The code does not panic and call everything unstable.

Recommended future refinement (beta-7): Expand classifier vocabulary to include `thermal_disorder`, `localized_relaxation`, `broad_disorder`, `atom_excess`, `atom_deficit`, `substitutional_site`. The numerical detectors are already stronger than the current labels.

6 Task 10 — Surface Interaction Analysis

PASS

Surface interaction behavior validated from trajectory only across six cases:

Key result: The stream case detected embedded behavior without storing `surface_damage` or `embedded` as state truth. Classification is derived entirely from trajectory displacement and contact duration.

Recommended future refinement (beta-7): Classifier priority should enforce event hierarchy:

Case	Observed classification
low normal velocity	no_interaction
medium normal velocity	no_interaction
grazing	no_interaction
angled impact	no_interaction
high normal velocity	temporary_residence / contact
stream injection	embedded_event

Table 5: Surface interaction classification results.

```
embedded_event > ejection_event > temporary_residence >
contact_no_capture > no_interaction
```

7 Task 12 / 12B — Diffusion and Macro Transport Inference

New file: `src/analysis/diffusion_analysis.hpp`

7.1 Architecture

Type	Purpose
DiffusionRecord	Per-frame: MSD total/per-axis, path length, tortuosity, jump count, residence time, neighbor exchange, KE proxy, energy drift, motion region
DiffusionTracker	Feeds position/velocity frames; computes all trajectory-derived metrics
TransportInference	Per-case macro transport inference table (analysis-only)
TransportAnalyzer	Fits MSD slope via linear least-squares; classifies transport regime
annotate_defect_ratios	Computes $D_{\text{ratio_vs_ideal}}$ by comparing defect cases to ideal reference
ActivationTrend::fit	Fits $\ln(D)$ vs $1/KE_{\text{proxy}}$ over multi-temperature cases
annotate_formation_context	Stamps formation-energy proxy and context label (analysis-only)

Table 6: Diffusion analysis module architecture.

7.2 Measured Quantities (per frame)

MSD total; MSD by axis (x, y, z); net displacement; path length; tortuosity proxy; jump count (site-to-site transitions); residence time; neighbor exchange count; kinetic energy proxy; energy drift (relative to baseline); motion region (surface / bulk).

7.3 Inferred Macro Transport Quantities (per case, analysis-only)

D_{eff} ($\text{\AA}^2/\text{fs}$ from MSD slope / $2d$); D_x, D_y, D_z ; anisotropy ratio = $\max(D)/\min_{\text{nonzero}}(D)$; jump rate; mean residence time; transport class; anisotropy class; D_{fit} quality (R^2); $D_{\text{ratio_vs_ideal}}$; E_{ref} , E_{atom} , ΔU proxy; formation context.

7.4 Test Results — 12 Diffusion Cases

All cases: NVE Lennard-Jones Argon ($Z = 18$, $\sigma = 3.4 \text{\AA}$, $\varepsilon = 0.238 \text{ kcal/mol}$, $m = 39.948 \text{ amu}$, $\Delta t = 2 \times 10^{-3} \text{ fs}$, $r_c = 10 \text{\AA}$).

7.4.1 SC 3×3×3 Crystal Cases (12 000 steps, seed = 77)

case_id	dim	MSD slope ($\text{\AA}^2/\text{fs}$)	D_{eff} ($\text{\AA}^2/\text{fs}$)	D_{ratio}	ΔU (kcal/mol)
case_ideal	3	0.039578	0.006596	1.000	0.0000
case_vacancy	3	0.239472	0.039912	6.051	+0.0603
case_interstitial	3	0.252350	0.042058	6.376	+34.631
case_mixed_defect	3	0.592508	0.098751	14.971	+0.1057

Table 7: SC defect diffusion cases.

7.4.2 Alternative Crystal Structures

case_id	dim	D_{eff} ($\text{\AA}^2/\text{fs}$)	E/atom (kcal/mol)	ΔU	context
case_fcc ($a = 5.26 \text{\AA}$)	3	0.030353	−1.2530	−0.5883	ideal_reference
case_bcc ($a = 3.30 \text{\AA}$)	3	0.025502	+11.2433	+11.9080	ideal_reference

Table 8: FCC and BCC structural comparison cases.

FCC is lower energy per atom than SC reference (−0.59 kcal/mol), consistent with LJ-parameterized Ar preferring close-packed structures. BCC at $a = 3.30 \text{\AA}$ is repulsive (+11.9 kcal/mol), correctly flagging over-compressed packing.

7.4.3 4-Point Thermal Activation Sweep (20 000 steps each)

case_id	σ_{jitter}	D_{eff} ($\text{\AA}^2/\text{fs}$)	$D_{\text{ratio_vs_ideal}}$
case_thermal_lo	0.05	0.061358	9.30×
case_thermal_med	0.12	0.367806	55.76×
case_thermal_hi	0.30	2.385689	361.67×
case_thermal_vhi	0.55	8.184049	1240.70×

Table 9: Thermal sweep: D_{eff} spans 130× across four cases.

D_{eff} spans 130× across the thermal sweep. The activation trend classifier returns `insufficient_data` because all NVE cases flag `invalid_energy_drift` — no thermostat, no PBC, energy drifts. This is correct. The flag gates the Arrhenius fit appropriately.

7.4.4 Reduced-Dimension Diffusion

Surface case: $D_z = 0.233$ vs $D_x = 0.112$ / $D_y = 0.117$. The out-of-plane axis shows higher apparent diffusivity, consistent with the unconstrained z -direction in a flat slab with no PBC.

7.5 Per-Axis Anisotropy Summary

8 Task 13 / 13B — Packing and Macro Property Inference

New file: `src/analysis/packing_analysis.hpp`

8.1 Architecture

8.2 Inferred Macro Packing Quantities (per case, analysis-only)

Bulk density proxy (occupied volume / container volume); bulk density inferred (mass-normalized); porosity = $1 - \phi$; mean coordination; largest cluster fraction; load-bearing network score; permeability proxy (Kozeny–Carman-style void/tortuosity estimate); sintering readiness proxy; packing macro class; compressibility proxy; densification rate proxy; contact growth rate.

case_id	dim	D_{eff} ($\text{\AA}^2/\text{fs}$)	anisotropy_class
case_surface (SC $5 \times 5 \times 1$ slab)	2	0.231231	anisotropic_transport
case_channel_1d (SC $1 \times 1 \times 20$ rod)	1	1.101889	weakly_anisotropic

Table 10: Reduced-dimension diffusion cases.

case_id	D_x	D_y	D_z	aniso ratio	class
case_ideal	0.00418	0.00813	0.00749	1.946	weakly_anisotropic
case_vacancy	0.02494	0.04679	0.04800	1.925	weakly_anisotropic
case_interstitial	0.03366	0.04761	0.04491	1.414	weakly_anisotropic
case_mixed_defect	0.06208	0.11249	0.12169	1.960	weakly_anisotropic
case_fcc	0.02812	0.02735	0.03559	1.302	weakly_anisotropic
case_bcc	0.03115	0.02751	0.01786	1.744	weakly_anisotropic
case_surface	0.11209	0.11688	0.23349	2.083	anisotropic_transport
case_channel_1d	0.27187	0.45973	0.37029	1.691	weakly_anisotropic

Table 11: Per-axis diffusivity and anisotropy classification. All values in $\text{\AA}^2/\text{fs}$.

8.3 Test Results — 4 Packing Cases

Container: $30 \times 30 \times 30 \text{\AA}^3$ ($27000 \text{\AA}^3$), bead radius $R = 1.7 \text{\AA}$.

Compressibility note: `densification_rate` = 0.000 (geometry unchanged between initial/-compressed frames in this test); `contact_growth_rate` = 5.143 contacts/frame. The compressibility proxy requires a physical pressure model to become non-trivial — labeled as a proxy because scientific honesty is cheaper than retractions.

9 Group 24 Test Binary

File	tests/test_macro_inference.cpp
CMake group	24
Target	test_macro_inference

Coverage:

- 12 diffusion cases (SC ideal, vacancy, interstitial, mixed-defect, FCC, BCC, thermal $\times 4$, surface 2D, channel 1D)
- 4 packing cases (loose, settled, compressed, jammed)
- Formation-context annotation per diffusion case
- Per-axis anisotropy audit table
- Formation coupling table ($E_{\text{ref}}/\text{atom}$, E/atom , ΔU , D_{eff})
- xyzFull audit assertions on every run

All pass conditions satisfied:

- D_{eff} computed from MSD slope only ✓
- Porosity computed from measured packing fraction only ✓
- No inferred property written to State ✓
- Energy drift cases correctly flagged ✓
- Dimensional reduction (2D, 1D) correctly applied ✓
- Formation context labels correct for all 12 cases ✓
- Thermal ordering: $D_{\text{vhi}} \geq D_{\text{lo}}$ ($130\times$ range confirmed) ✓

Type	Purpose
PackingRecord	Per-frame: packing fraction, coordination, largest cluster, wall proxy, contact persistence, void connectivity, settling flag, energy drift
PackingTracker	$O(N^2)$ contact network from bead geometry; BFS for largest connected cluster
PackingInference	Per-case macro packing table (analysis-only)
PackingAnalyzer	Infers all macro properties from geometry/contact evolution
compute_compressibility	Static; takes initial + compressed snapshots, derives $\Delta U/\Delta V$ proxies

Table 12: Packing analysis module architecture.

case_id	N	ϕ	porosity	$\bar{z}$	cluster	perm.	sinter.	class
pack_loose	50	0.0381	0.9619	0.56	0.060	612.76	0.055	loose_powder
pack_settled	343	0.2614	0.7386	0.018	0.006	5.894	0.001	loose_powder
pack_compressed	343	0.2614	0.7386	5.125	1.000	5.894	0.933	sintering_candidate
pack_jammed	512	0.3902	0.6098	5.211	1.000	1.489	0.933	sintering_candidate

Table 13: Packing inference results. ϕ = packing fraction; $\bar{z}$ = mean coordination; cluster = largest cluster fraction; perm. = permeability proxy; sinter. = sintering readiness proxy.

10 Formation Codebase — Existing Infrastructure Survey

The following formation infrastructure exists and is ready for beta-7 wiring:

Integration path for 12B $\leftrightarrow$ Formation (beta-7 target):

```
trajectory
  -> DiffusionTracker -> TransportInference
  + annotate_formation_context(ref_energy_per_atom from UnitCell preset
    )
  + ThermoDatabase::get(formula) -> H_f for true delta-H comparison
  + cluster_by_defect(snapshots) -> microstate classification
  -> formation_energy_proxy bridges D_eff <-> H_f
```

The `formation_energy_proxy` field now present in `TransportInference` is the first physical bridge between the trajectory analysis layer and the thermodynamic formation layer. It is a ΔU proxy, not a true DFT formation enthalpy. It is labeled accordingly.

11 Known Limitations and Honest Notes

11.1 Energy Drift in NVE Lennard-Jones Tests

All 12 diffusion cases flag `invalid_energy_drift` and `valid_for_macro_inference = no`. This is correct. NVE integration without PBC or thermostat on a small SC crystal drifts. The flag works. D_{eff} values are still computed and reported — they reflect kinetic behavior, not thermodynamic equilibrium.

For physically valid D_{eff} values, the system requires a thermostat (Langevin already exists in `actions_form.cpp`) and PBC (supported in `atomistic::Box`). That wiring is a beta-7 task.

11.2 BCC Argon at $a = 3.30 \text{ \AA}$

$\Delta U = +11.9 \text{ kcal/mol}$ vs SC reference because $a = 3.30 \text{ \AA}$ is well inside the LJ repulsive wall for Ar ($\sigma = 3.4 \text{ \AA}$). The BCC test correctly shows overcompressed-repulsive behavior. It is not

Layer	File	Key symbols
Formation loop (MD)	src/cli/actions_form.cpp	LangevinDynamics::integrate(), T-schedule, write_snapshot, formation.log
Formation priors	apps/phase4_formation_priors.cpp	crystal preset → UnitCell → to_state() → single-point energy
Thermodynamics	include/thermo/thermodynamics.hpp	ThermoData {H_f, S, G_f, Cp}, ThermoDatabase, GibbsCalculator::calculate(), estimate_H_formation()
Fingerprinting	atomistic/classify/fingerprints.hpp	ProtoFingerprint, DefectFingerprint, weisfeiler_lehman_hash()
Structure clustering	atomistic/classify/cluster.hpp	cluster_by_proto(), cluster_by_defect(), polymorph/isomor- ph/defect classify
Potential energy	src/pot/energy.hpp	EnergyResult {UvdW, UCoul, total}, bond/an- gle/torsion params
Formation regime records	tests/test_runners.hpp	FormationRegimeRecord, FormationHistory, build_regime_record(), relaxation_time()

Table 14: Formation infrastructure ready for beta-7 wiring.

a physical BCC Ar ground state. The label `ideal_reference` refers to the geometric reference classification, not a claim of thermodynamic stability.

11.3 Interstitial $\Delta U = +34.6$ kcal/mol

The interstitial atom is placed at an off-lattice position in the repulsive core of its neighbors. The large ΔU correctly reflects that a naive geometric interstitial injection is high-energy. A FIRE-relaxed interstitial would show a smaller and physically meaningful formation energy. FIRE relaxation is available in the existing integrator stack and is a beta-7 enhancement target.

11.4 Activation Trend Returns `insufficient_data`

The 4-point thermal sweep produces a $130\times D$ range but all cases are `invalid_energy_drift`. The Arrhenius fit gate correctly excludes them. To get a real activation trend, the cases must be thermostatted. The infrastructure is correct. The physics input is not yet calibrated.

12 Beta Release Definitions

13 Executive Summary

Beta-6 completes the instrumentation layer of VSEPR-SIM v5. The project now has:

Version	Label	What ships
v4.1.0	Stable foundation	Frozen known-good baseline
v5.0.0-beta.6	Formation and Premacro Analysis Foundation	Formation loop, thermodynamics, fingerprinting, clustering, crystal imperfection, surface analysis, diffusion inference, packing inference, formation-context annotation
v5.0.0-beta.7	Formation/Analysis Unification and Report Freeze	Connect formation outputs $\rightarrow$ fingerprint $\rightarrow$ cluster $\rightarrow$ analyze $\rightarrow$ report
v5.0.0-beta.8	Coupled Premacro-to-Macro Workflow Prototype	Material Candidate Card, imperfection $\rightarrow$ diffusion workflow, packing $\rightarrow$ macro workflow, candidate-level screening

Table 15: Beta release roadmap.

- a formation engine skeleton (formation loop, priors, thermodynamics, fingerprinting, clustering, regime records)
- a premacro measurement backbone (Vec3 authority, Eigen bridge, Kabsch/RMSD, stationarity, crystal imperfection, surface interaction)
- trajectory-to-inference pipelines for diffusion and packing
- analysis-only macro property inference with explicit state-truth separation
- formation-context annotation bridging ΔU proxies to transport behavior
- 12 validated diffusion cases and 4 validated packing cases with exact numerical outputs

The architecture rule held. No inferred macro property is stored in State.

Beta-6 is the instrumentation release. Beta-7 wires the instruments together. Beta-8 uses them for screening.

The formation codebase survey printed in `test_macro_inference` output documents the precise integration path forward.

Report generated from live test run. All numerical values confirmed reproducible across multiple executions.

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